

Electromagnetic Waves

Static Electric Fields



Fundamental laws:

- Coulomb's Law
- Gauss's Law

(both these laws are based on experimental studies and are interdependent)

Fundamental Postulates of Electrostatics in Free Space



Electric field intensity

$$\mathbf{E} = \lim_{q \rightarrow 0} \frac{\mathbf{F}}{q} \quad (\text{V/m}).$$

Inverse Relation:

$$\mathbf{F} = q\mathbf{E} \quad (\text{N}).$$

Fundamental Postulates:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \times \mathbf{E} = 0$$

Fundamental Postulates of Electrostatics in Free Space



Fundamental Postulates:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \times \mathbf{E} = 0$$

$$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$$

Gauss's Law

$$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0$$

Total outward flux of the E-field over any closed surface in free space is equal to the total (net) charge enclosed in the surface divided by ϵ_0 .

The scalar line integral of the static electric field intensity around any closed path vanishes

Fundamental Postulates of Electrostatics in Free Space



Summary

Postulates of Electrostatics in Free Space	
Differential Form	Integral Form
$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$	$\oint_S \mathbf{E} \cdot d\mathbf{s} = \frac{Q}{\epsilon_0}$
$\nabla \times \mathbf{E} = 0$	$\oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0$

Fundamental Postulates of Magnetostatics in Free Space



A magnetostatic field is produced by a constant current flow.

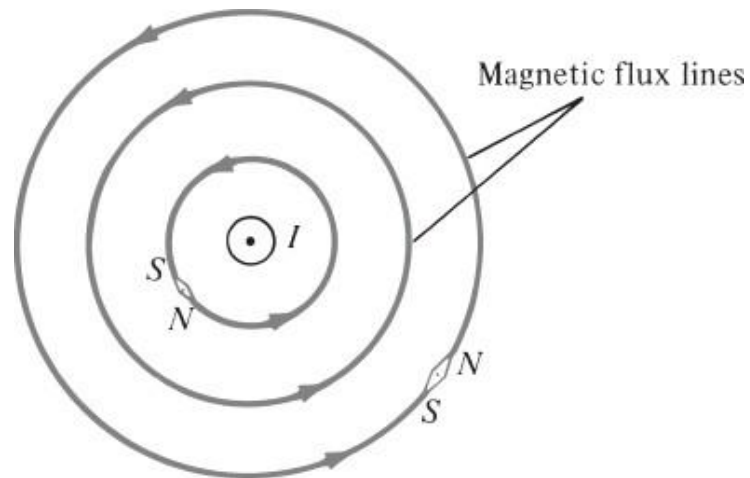
1. Divergence of $\mathbf{B}$

2. Curl of $\mathbf{B}$

Fundamental Postulates of Magnetostatics in Free Space



$$\psi = \int_s B \cdot dS$$



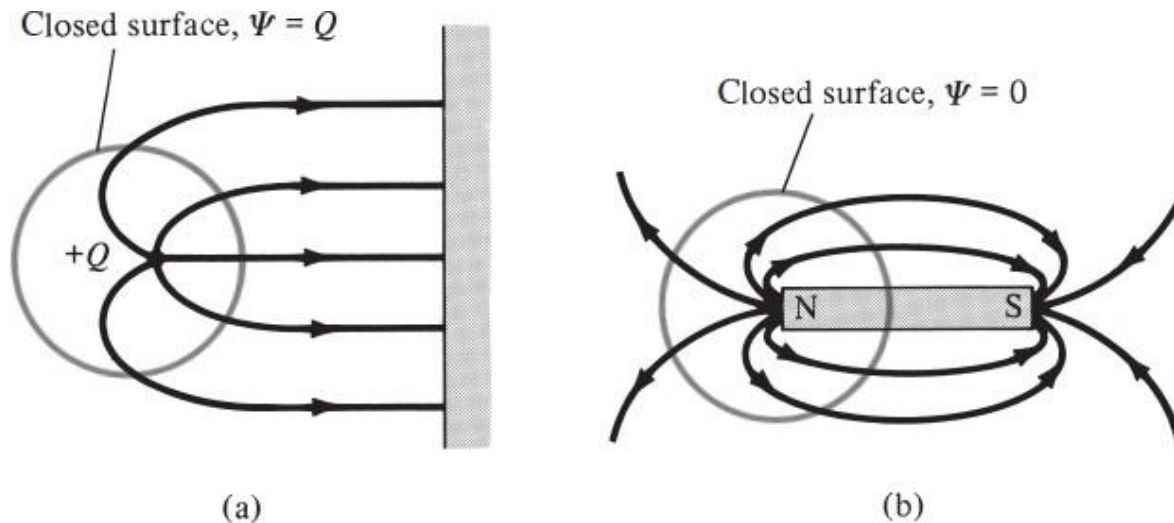


Figure 6.17 Flux leaving a closed surface due to (a) isolated electric charge $\Psi = \oint_S \mathbf{D} \cdot d\mathbf{S} = Q$, (b) magnetic charge, $\Psi = \oint_S \mathbf{B} \cdot d\mathbf{S} = 0$.

$$\oint_c \vec{B} \cdot d\vec{S} = 0$$
$$\vec{\nabla} \cdot \vec{B} = 0$$

Gauss's Law for magnetic fields

It is not possible to have isolated magnetic poles

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

Tell us try to have its integral form



Ampere's Circuital Law

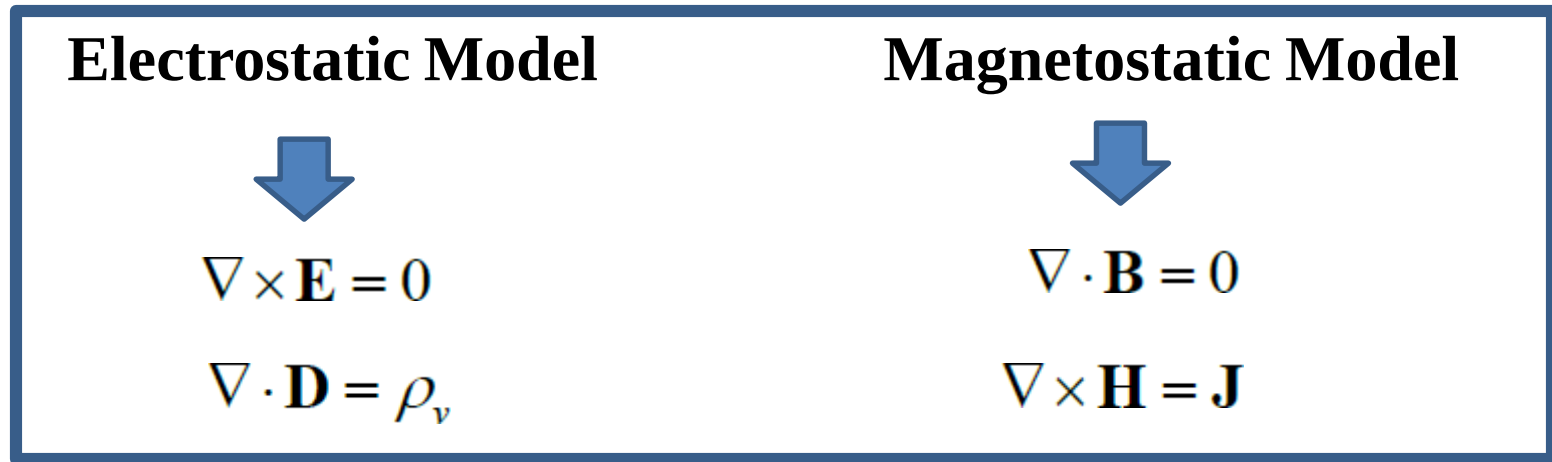
Ampere's circuital law states that the line integral of $\mathbf{H}$ around a closed path is the same as the net current enclosed by the path.

Magnetostatic Field is not conservative

Maxwell's Equation for **Static** electric field and **steady** magnetic field

Differential (or Point) Form	Integral Form	Remarks
1. $\nabla \cdot \mathbf{D} = \rho_v$	$\oint_s \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv$	Gauss's Law
2. $\nabla \cdot \mathbf{B} = 0$	$\oint_s \mathbf{B} \cdot d\mathbf{S} = 0$	Nonexistence of magnetic monopole
3. $\nabla \times \mathbf{E} = 0$	$\oint_s \mathbf{E} \cdot d\mathbf{l} = 0$	Conservative nature of electrostatic field.
4. $\nabla \times \mathbf{H} = \mathbf{J}$	$\oint_s \mathbf{H} \cdot d\mathbf{l} = \int_s \mathbf{J} \cdot d\mathbf{S}$	Ampere's Law

Maxwell's equations (Time Varying)



Static EM fields, electric and magnetic fields are **independent** of each other

However

Dynamic EM fields, electric and magnetic fields are **interdependent** of each other



Electromagnetic Model

Faraday's Law of Electromagnetic Induction



Experimentally discovered: A current was induced in a conducting loop when the magnetic flux linking the loop changed.

Fundamental Postulate for Electromagnetic Induction

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

It says that the electric field intensity in a region of time-varying magnetic flux density is therefore non-conservative and cannot be expressed as the gradient of a scalar potential.

Maxwell's Equations



$$\nabla \times \mathbf{H} = \mathbf{J}$$

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J}$$

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \neq 0$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_d$$

Concept of Displacement Current:

Maxwell's Equations in Time-varying Fields



<u>Differential</u>	<u>Integral</u>	<u>Remarks</u>
1 $\nabla \cdot D = \rho_v$	$\oint_S D \cdot dS = \int_V \rho_v dv$	<u>Gauss' Law</u>
2 $\nabla \cdot B = 0$	$\oint_S B \cdot dS = 0$	<u>Nonexistence of isolated charge</u>
3 $\nabla \times E = -\frac{\partial B}{\partial t}$	$\oint_L E \cdot dl = -\frac{\partial}{\partial t} \int_S B \cdot dS$	<u>Faraday's Law</u>
4 $\nabla \times H = J + \frac{\partial D}{\partial t}$	$\oint_L H \cdot dl = \int_S \left(J + \frac{\partial D}{\partial t} \right) \cdot dS$	<u>Modified Ampere's law</u>

In 1 and 2, S is a closed surface enclosing the volume V

In 3 and 4, L is a closed path that bounds the surface S

Concept of Displacement Current:

Maxwell's Equations in Time-varying Fields



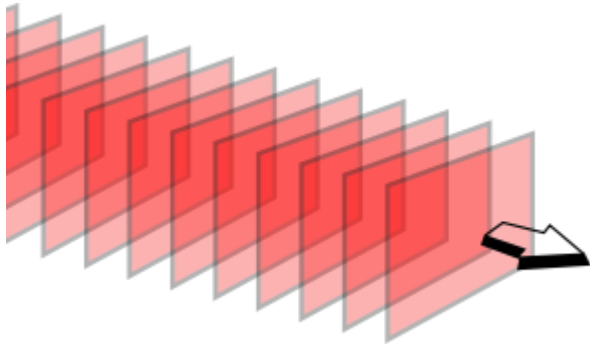
Uniform Plane Wave



Uniform Plane Wave



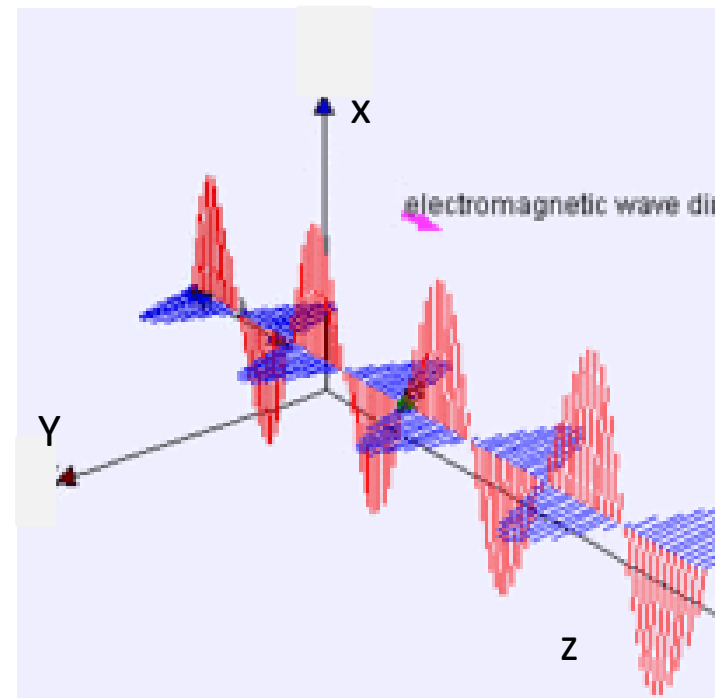
Electric Field (and also $\mathbf{H}$) assumed to have same direction, same magnitude, and same phase in infinite planes perpendicular to the direction of propagation.



The wavefronts of a plane wave traveling in space

$$\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0,$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \cong 3 \times 10^8 \text{ (m/s)} = 300 \text{ (Mm/s)}$$



Plane Waves in Lossless Media



$$\nabla^2 \mathbf{E} + k_0^2 \mathbf{E} = 0,$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k_0^2 \right) E_x = 0.$$

$$\frac{d^2 E_x}{dz^2} + k_0^2 E_x = 0,$$

$$\begin{aligned} E_x(z) &= E_x^+(z) + E_x^-(z) \\ &= E_0^+ e^{-jk_0 z} + E_0^- e^{jk_0 z} \end{aligned}$$

$$\begin{aligned} E_x^+(z, t) &= \Re[E_x^+(z) e^{j\omega t}] \\ &= \Re[E_0^+ e^{j(\omega t - k_0 z)}] \\ &= E_0^+ \cos(\omega t - k_0 z) \quad (\text{V/m}). \end{aligned}$$

$$k_0 = \frac{2\pi}{\lambda_0} \quad (\text{rad/m}),$$

$\omega t - k_0 z = \text{A constant phase,}$

$$\frac{dz}{dt} = \frac{\omega}{k_0} = c.$$

Plane Waves in Lossless Media



Considering the source free wave equation for free space:

$$\nabla^2 \mathbf{E} + k_0^2 \mathbf{E} = 0,$$

$$k_0 = \omega \sqrt{\mu_0 \epsilon_0} = \frac{\omega}{c} \quad (\text{rad/m}).$$

Traveling Wave

